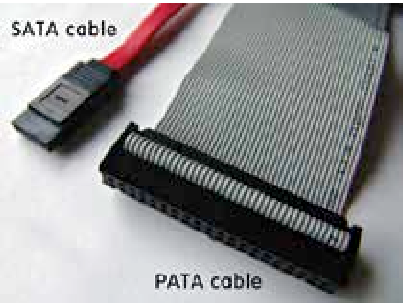
about inertia in your physics class, this should make sense to you). So it is important to look at the power consumption of each disk you want to add and the rating of your power supply unit. Also, look for the interface of the HDD. SATA is the new interface while IDE/PATA is the old one. Nowadays, it would be hard to come across a PATA/IDE interface, but we advise you to pay attention regardless. It's also worth remembering that an E-SATA is not the same as SATA. E-SATA stands for External SATA and is not a compatible interface with SATA. E-SATA disks are meant to be used as external disks (given there is a port on the motherboard or front panel of the cabinet). Also, do not underestimate the power of a decent amount of cache within the hard disk. They can affect the performance to a good extent for certain types of workloads, especially when you need to read fragmented files back-and-forth. The hard disk can serve those blocks right from the cache, if the space allows it.



SATA-vs-PATA

In case it wasn't obvious, you should also check for the size of the hard-disk (2.5 or 3.5-inch). The larger ones (3.5-inches) are meant for desktops while the smaller ones (2.5-inches) are made for laptops.

Why SSDs are fast

SSDs employ flash technology (mostly NAND flash) to achieve high speeds. It is worth remembering that each flash memory chip has a limited number of write operations that it can take. Let's call it the write threshold. After you have written more than that many number of times on a single area i.e. after you've crossed the write threshold, that particular area on the flash memory degrades and the data stored on it may get corrupted. Hence, it is recommended to keep data which is written less often as compared to the number of times it's read on SSDs.

Also, filling the disk with data up to its upper limit will reduce the physical area available for writing while the already resident data covers most of the storage area. Don't let this piece of information drive you paranoid though. Most SSDs have enough read-write cycles to last you for years.